EEE-2421: Electrical Drives and Instrumentation

Credit Hours - 2 **Segment - 1**

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What is Electrical Machine?

- ✓ An electrical machine is a device that can convert either mechanical energy to electrical energy or electrical energy to mechanical energy.

- ✓ When such a device isused to convert mechanical energy to electrical energy, it is called a generator.

- ✓ When it converts electrical energy to mechanical energy, it is called a motor.

- ✓ Transformer is another type of machine to accomplish the change in voltage level.

Basic concept of electrical machines fundamentals

- ✓ Angular position : The angular position θ of an object is the angle at which it is oriented, measured from some arbitrary reference point/axis. Measuring unit: radians or degrees.

- ✓ Angular Velocity : Angular velocity (or speed) is the rate of change in angular position with respect to time.

$$\omega = \frac{d\theta}{dt}$$

Measuring unit: rad/sec, rpm.

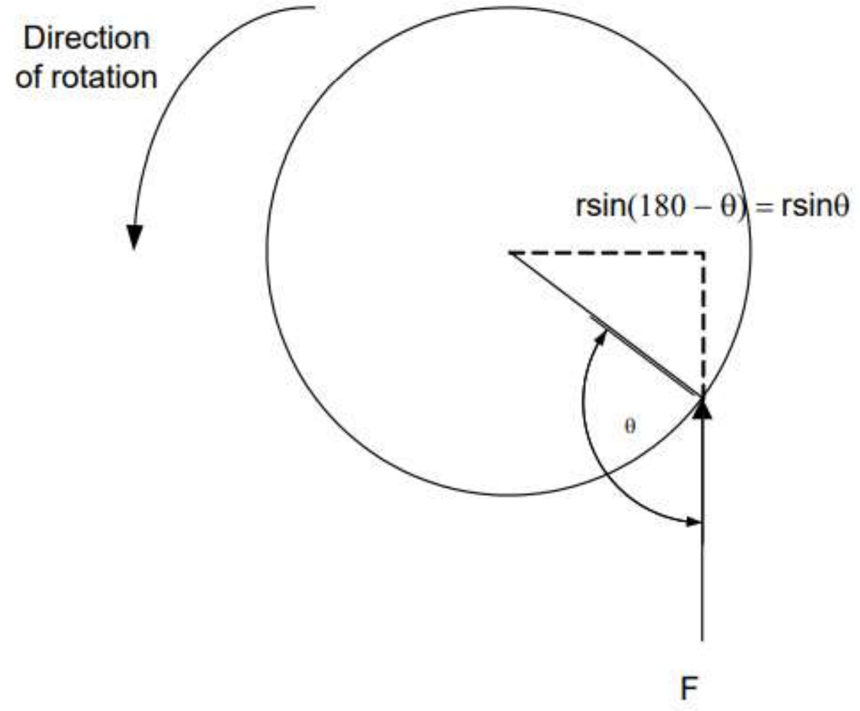
Basic concept of electrical machines fundamentals

- ✓ Angular acceleration : Angular acceleration is the rate of change in angular velocity with respect to time. Measuring unit: rad/sec^2

$$\alpha = \frac{d\omega}{dt}$$

- ✓ Torque : Torque is the measure of the force that can cause an object to rotate about an axis.

$$\tau = \text{Force} \times \text{perpendicular distance}$$
$$= F \times r \sin\theta$$



Basic concept of electrical machines fundamentals

✓ Work

For linear motion, work is defined as the application of a *force* through a *distance*. In equation form,

$$W = \int F\,dr \tag{1-9}$$

For rotational motion, work is the application of a *torque* through an *angle*. Here the equation for work is

$$W = \int \tau\,d\theta \tag{1-11}$$

✓ Power

Power is the rate of doing work, or the increase in work per unit time. The equation for power is

$$P = \frac{dW}{dt} \tag{1-13}$$

By this definition, and assuming that force is constant and collinear with the direction of motion, power is given by

$$P = \frac{dW}{dt} = \frac{d}{dt}(Fr) = F\left(\frac{dr}{dt}\right) = Fv \tag{1-14}$$

Similarly, assuming constant torque, power in rotational motion is given by

$$P = \frac{dW}{dt} = \frac{d}{dt}(\tau\theta) = \tau\left(\frac{d\theta}{dt}\right) = \tau\omega$$

$$P = \tau\omega \tag{1-15}$$

Equation (1-15) is very important in the study of electric machinery, because it can describe the mechanical power on the shaft of a motor or generator.

Basic concept of electrical machines fundamentals

✓ Newton's Law of Rotation

Newton's law for objects moving in a straight line gives a relationship between the force applied to the object and the acceleration experience by the object as the result of force applied to it. In general,

$$F = ma$$

where:

F – Force applied
m – mass of object
a – resultant acceleration of object

Applying these concept for rotating bodies,

$$\tau = J\alpha \text{ (Nm)}$$

where:

τ - Torque
J – moment of inertia
α - angular acceleration

The Magnetic Field

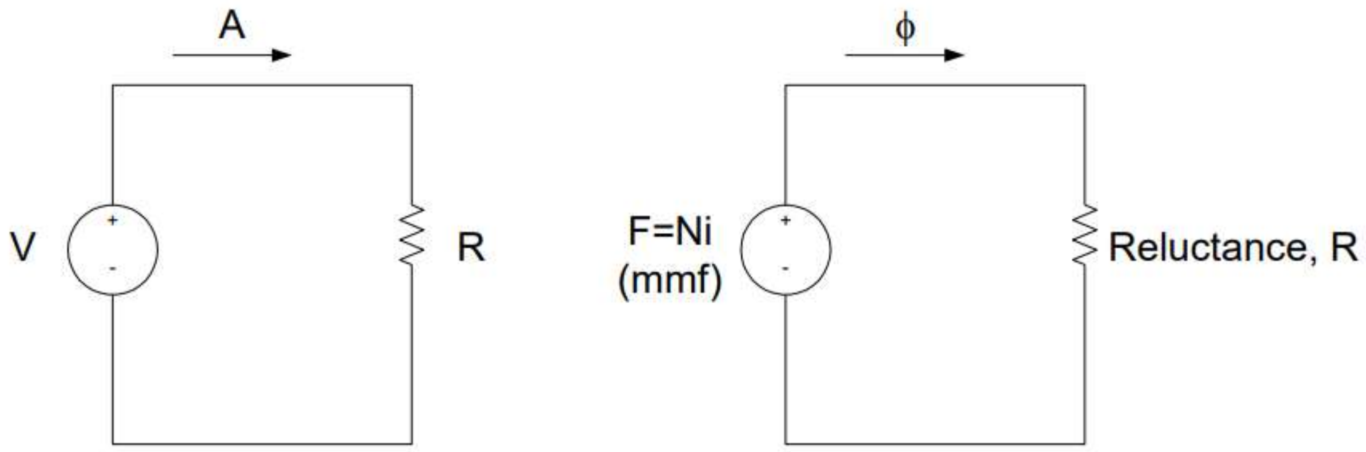
- ✓ Magnetic fields are the fundamental mechanism by which energy is converted from one form to another in motors, generators, and transformers.

- ✓ Four basic principles describe how magnetic fields are used in these devices:

1. A current-carrying wire produces a magnetic field in the area around it.
2. A time-changing magnetic field induces a voltage in a coil of wire if it passes through that coil. (This is the basis of transformer action.)
3. A current-carrying wire in the presence of a magnetic field has a force induced on it. (This is the basis of motor action.)
4. A moving wire in the presence of a magnetic field has a voltage induced in it. (This is the basis of generator action.)

Magnetic Circuit

- ✓ The flow of magnetic flux induced in the ferromagnetic core can be made analogous to an electrical circuit hence the name magnetic circuit.



Electric Circuit Analogy *Magnetic Circuit Analogy*

Magnetic Behaviour of Ferromagnetic Materials

- ✓ **Non-magnetic materials** show a linear relationship between the flux density B and coil current I. In other words, they **have constant permeability. But in iron and other ferromagnetic materials it is not constant.**

- ✓ The permeability is the property of a medium that determines its magnetic characteristics.

- ✓ In electrical machines and electromechanical devices a somewhat linear relationship between B and I is desired, which is normally approached by limiting the current.

Magnetic Behaviour of Ferromagnetic Materials

- ✓ **Faraday's Law :** If a flux passes through a turn of coil of wire, voltage will be induced in the turn of the wire that is directly proportional to the rate of change in the flux with respect of time.

$$\boxed{e_{ind} = -N\frac{d\phi}{dt}}$$

- ✓ **Induced Force on a Wire :** A current carrying conductor present in a uniform magnetic field of flux density B, would produce a force to the conductor/wire. The force induced is given by:

$$\boxed{F = ilB\sin\theta}$$

where:
i – represents the current flow in the conductor
l – length of wire, with direction of l defined to be in the direction of current flow
B – magnetic field density

Magnetic Behaviour of Ferromagnetic Materials

✓Induced Voltage on a Conductor Moving in a Magnetic Field

$$\boxed{e_{ind} = (v \times B) l \cos\theta}$$

where:

v – velocity of the wire
B – magnetic field density
l – length of the wire in the magnetic field
θ - angle between the conductor and the direction of (***v x B***)

Related Problems

Example 1–6. Figure 1–15 shows a coil of wire wrapped around an iron core. If the flux in the core is given by the equation

$$\phi = 0.05 \sin 377t \qquad \text{Wb}$$

If there are 100 turns on the core, what voltage is produced at the terminals of the coil?

Solution

By the same reasoning as in the discussion on pages 29–30, the direction of the voltage while the flux is increasing in the reference direction must be positive to negative, as shown in Figure 1–15. The *magnitude* of the voltage is given by

$$e_{\text{ind}} = N\frac{d\phi}{dt}$$

$$= (100 \text{ turns})\frac{d}{dt}(0.05 \sin 377t)$$

$$= 1885 \cos 377t$$

or alternatively,

$$e_{\text{ind}} = 1885 \sin(377t + 90°) \text{ V}$$

Example 1–7. Figure 1–16 shows a wire carrying a current in the presence of a magnetic field. The magnetic flux density is 0.25 T, directed into the page. If the wire is 1.0 m long and carries 0.5 A of current in the direction from the top of the page to the bottom of the page, what are the magnitude and direction of the force induced on the wire?

Solution

The direction of the force is given by the right-hand rule as being to the right. The magnitude is given by

$$F = ilB \sin \theta \tag{1–44}$$

$$= (0.5 \text{ A})(1.0 \text{ m})(0.25 \text{ T}) \sin 90° = 0.125 \text{ N}$$

Therefore,

$$\mathbf{F} = 0.125 \text{ N, directed to the right}$$

The induction of a force in a wire by a current in the presence of a magnetic field is the basis of *motor action*. Almost every type of motor depends on this basic principle for the forces and torques which make it move.

Related Problems

Example 1–8. Figure 1–17 shows a conductor moving with a velocity of 5.0 m/s to the right in the presence of a magnetic field. The flux density is 0.5 T into the page, and the wire is 1.0 m in length, oriented as shown. What are the magnitude and polarity of the resulting induced voltage?

Solution

The direction of the quantity $\mathbf{v} \times \mathbf{B}$ in this example is up. Therefore, the voltage on the conductor will be built up positive at the top with respect to the bottom of the wire. The direction of vector $\mathbf{l}$ is up, so that it makes the smallest angle with respect to the vector $\mathbf{v} \times \mathbf{B}$.

Since $\mathbf{v}$ is perpendicular to $\mathbf{B}$ and since $\mathbf{v} \times \mathbf{B}$ is parallel to $\mathbf{l}$, the magnitude of the induced voltage reduces to

$$\begin{aligned} e_{\text{ind}} &= (\mathbf{v} \times \mathbf{B}) \bullet \mathbf{l} \qquad (1\text{–}45) \\ &= (vB \sin 90°)\, l \cos 0° \\ &= vBl \\ &= (5.0 \text{ m/s})(0.5 \text{ T})(1.0 \text{ m}) \\ &= 2.5 \text{ V} \end{aligned}$$

Thus the induced voltage is 2.5 V, positive at the top of the wire.

Related Problems

Example: A coil that has 700 turns develops an average induced voltage of 50 V. What must be the change in the magnetic flux that occur to produce such a voltage if the time interval for this change is 0.7 seconds?

Solution:

Given:

Number of Turn (N) = 700

Induced Voltage (ε) = 50 V

Time Interval (dt) = 0.7 s

By using Faraday's Law,

$\varepsilon = N\ (d\varphi/dt)$

Therefore, Change in flux (dφ) is,

$d\varphi = \varepsilon \times dt / N$

$\quad = 50 \times 0.7 / 700$

$d\varphi = 0.05\ Wb$

Example: The magnetic flux linked with a coil having 250 turns is changed from 1.4 Wb to 2 Wb in 0.45 seconds. Calculate the induced emf in the coil.

Solution:

Given:

Number or Turns (N) = 250

Time Interval (dt) = 0.45 s

Initial Flux $(\varphi_1) = 1.4\ Wb$

Final flux $(\varphi_2) = 2\ Wb$

Therefore, change in flux (dφ) can be given by,

$d\varphi = \varphi_2 - \varphi_1 = 2 - 1.4 = 0.6\ Wb$

According to Faraday's Law,

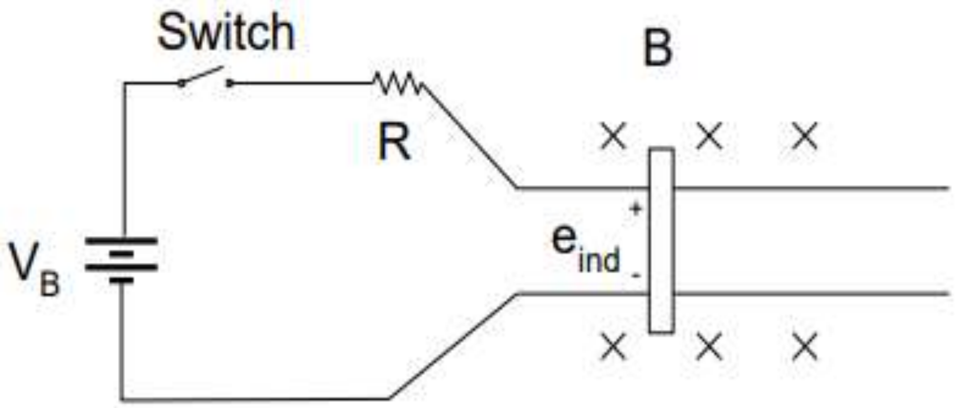
Induced Emf (ε) = N (dφ/dt)

$\varepsilon = 250 \times (0.6 / 0.45)$

$\varepsilon = 333.33\ V$

Linear DC Machine

Linear DC machine is the simplest form of DC machine which is easy to understand and it operates according to the same principles and exhibits the same behaviour as motors and generators.



Production of Force on a current carrying conductor

$$F = i(l \times B)$$

Voltage induced on a current carrying conductor moving in a magnetic field

$$e_{ind} = (v \times B)\, l$$

Kirchoff's voltage law

$$V_B - iR - e_{ind} = 0$$

$$\therefore V_B = e_{ind} + iR = 0$$

Newton's Law for motion

$$F_{net} = ma$$

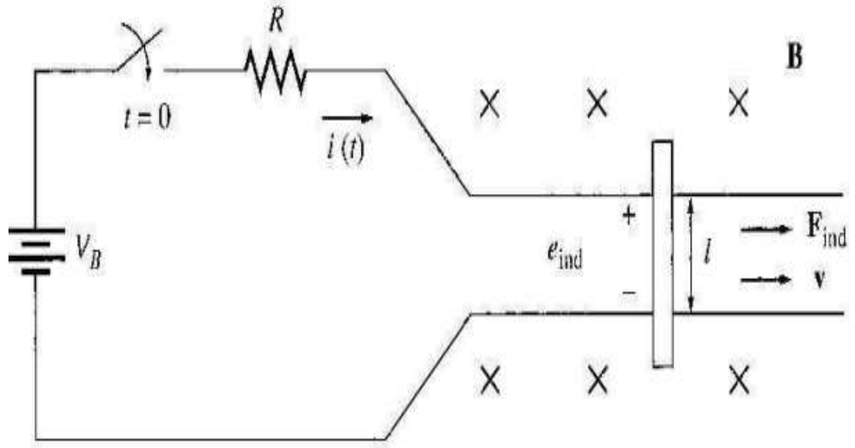
Starting the Linear DC Machine

The switch is closed and current will flow in the circuit.
At this moment, the induced voltage is 0 due to no movement of the wire (the bar is at rest).

$$Since, V_B = iR + e_{ind}$$

$$\therefore \quad i = \frac{V_B - e_{ind}}{R}$$

assuming that R is very small



As the current flows down through the bar, a force will be induced on the bar. (current flowing through a wire in the presence of a magnetic field induces a force in the wire)

$$F = i\,(l \times B) \\ = ilB \sin 90 \\ = ilB$$

Direction of movement:
Right

When the bar starts to move, its velocity will increase, and a voltage appears across the bar.

$$e_{ind} = (v \times B)l \\ = vBl \sin 90° \\ = ilB$$

Direction of induced potential: positive upwards

Due to the presence of motion and induced potential, the current flowing in the bar will reduce.
The result of this action is that eventually the bar will reach a constant steady-state speed where the net force on the bar is zero. This occurs when eind has risen all the way up to equal VB. This is given by:

$$V_B = e_{ind} = v_{steady\ state} Bl$$

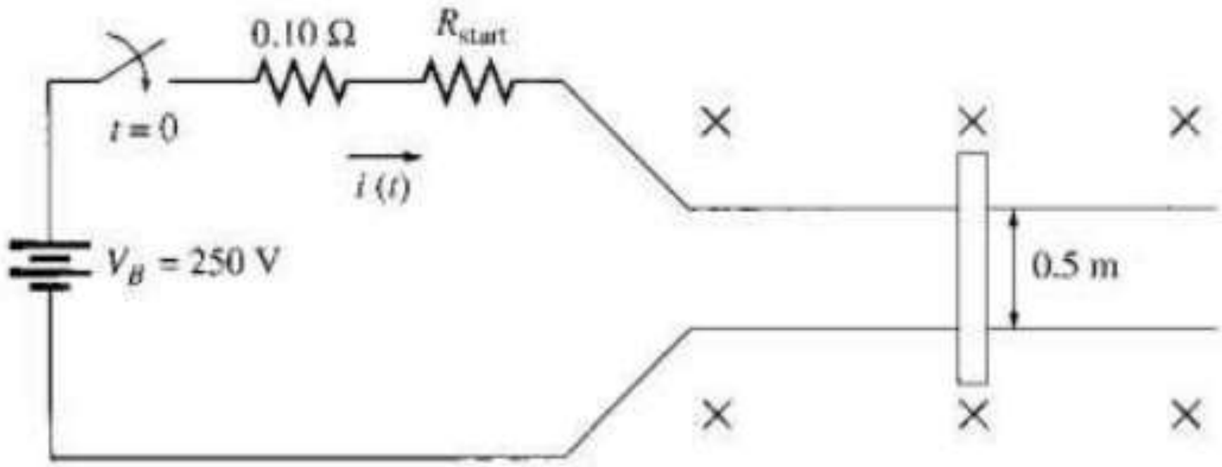
$$\therefore v_{steady\ state} = \frac{V_B}{Bl}$$

Starting Problems in Linear DC Machine

At starting, the speed of the bar is zero.
so eind = 0.
Resistance being so small, this current is very high (10x in excess of the rated current).
Prevention:
insert an extra resistance into the circuit during starting to limit current flow until eind builds up enough to limit it, as shown here:

$$\text{Since}, V_B = iR + e_{ind}$$

$$\therefore \quad i = \frac{V_B - e_{ind}}{R}$$



Linear DC Machine as Motor

Assume the linear machine is initially running at the no-load steady state condition and what happen when an external load is applied?

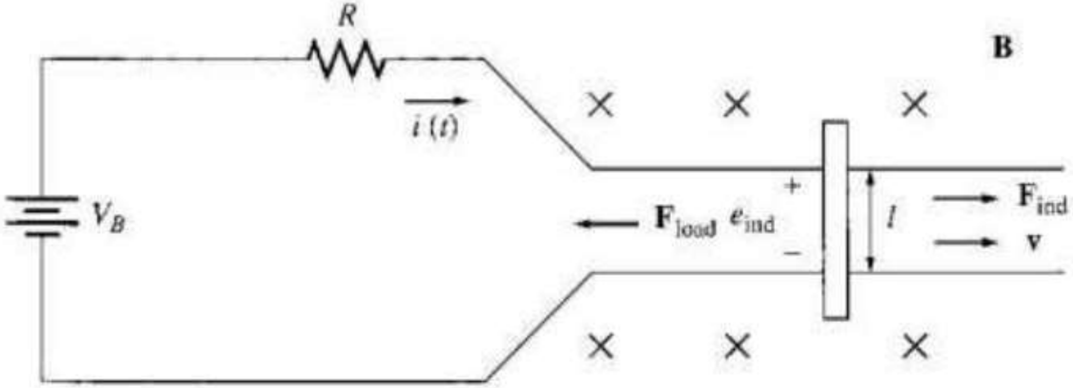
A force F_{load} is applied to the bar opposing the direction of motion. Since the bar was initially at steady state, application of the force F_{load} will result in a net force on the bar in the direction opposite the direction of motion.

$$F_{net} = F_{load} - F_{ind}$$

Thus, the bar will slow down (the resulting acceleration $a = F_{net}/m$ is negative). As soon as that happen, the induced voltage on the bar drops ($e_{ind} = v\downarrow Bl$).

When the induced voltage drops, the current flow in the bar will rise:

$$i\uparrow = \frac{V_B - e_{ind}\downarrow}{R}$$



Thus, the induced force will rise too. ($F_{ind}\uparrow = i\uparrow lB$)

Final result → the induced force will rise until it is equal and opposite to the load force, and the bar again travels in steady state condition, but at a lower speed. See graphs below:

Linear DC Machine as Motor

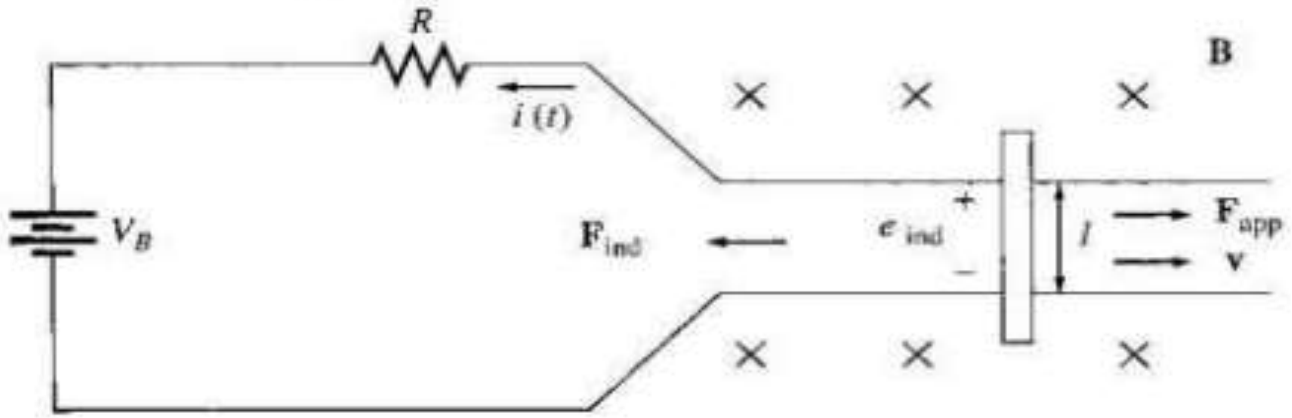
Now, there is an induced force in the direction of motion and power is being converted from **electrical to mechanical form** to keep the bar moving.

The power converted is $P_{conv} = e_{ind} I = F_{ind} v$ → An amount of electric power equal to $e_{ind} i$ is consumed and is replaced by the mechanical power $F_{ind} v$ → **MOTOR**

The power converted in a real rotating motor is: $P_{conv} = \tau_{ind} \omega$

Linear DC Machine as Generator

Assume the linear machine is operating under no-load steady-state condition. A force in the direction of motion is applied.



The applied force will cause the bar to accelerate in the direction of motion, and the velocity v will increase.

When the velocity increase, $e_{ind} = V \uparrow Bl$ will increase and will be larger than V_B.

When $e_{ind} > V_B$ the current reverses direction.

Since the current now flows up through the bar, it induces a force in the bar ($F_{ind} = ilB$ to the left). This induced force opposes the applied force on the bar.

End result → the induced force will be equal and opposite to the applied force, and the bar will move at a higher speed than before. The linear machine no is converting mechanical power $F_{ind}\,v$ to electrical power $e_{ind}\,i$ → GENERATOR

The amount of power converted : $P_{conv} = \tau_{ind}\,\omega$

Linear DC Machine as Motor or Generator

- The same machine acts as both motor and generator. The only difference is whether the externally applied force is in the direction of motion (generator) or opposite to the direction of motion (motor).
- Electrically, $e_{ind} > V_B \rightarrow$ generator
- $e_{ind} < V_B \rightarrow$ motor

Motor Parameters

Speed: The number of revolutions per minute of the motor. Denoted by N. Academically denoted by, $\omega = 2\pi N/60$. measured in rpm (revs per minute) or rps (revs per second)

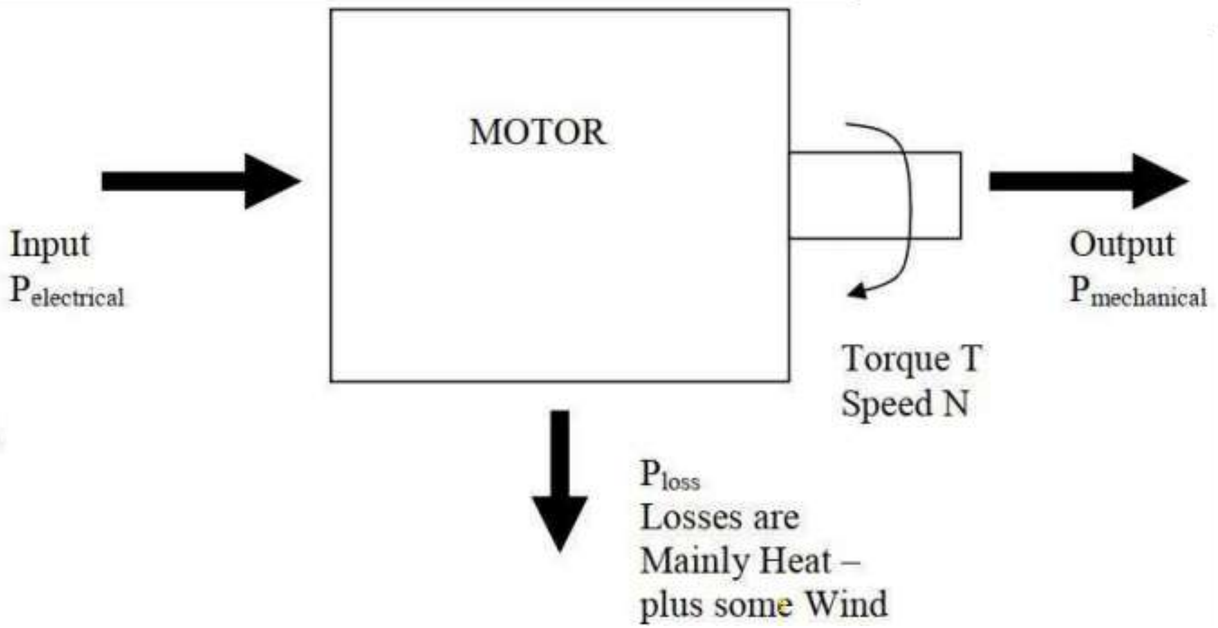
Torque: The force with which a motor turns is called Torque measured in Newton meters (Nm). Amotor with highTorque accelerates quickly and candrive aheavy load.

The relation between Power, Torque and Speed is given by:

$$P = \frac{2 \times \pi \times T \times N}{60}$$

$$RMS\,power = \sqrt{\frac{P_1^2 t_1 + P_2^2 t_2 + P_3^2 t_3 + \ldots.. + P_n^2 t_n}{t_1 + t_2 + t_3 + \ldots.. + t_n}}$$

Motor Losses



- Losses and Efficiency

$$Efficiency = \eta = \frac{Output}{Input} \times 100\%$$

$$Input = Output + Losses$$

$$P_{electrical} = P_{mechanical} + P_{loss}$$

$$\eta = \frac{P_{mechanical}}{P_{electrical}} \times 100\%$$

Eddy current loss

The power losses in conductors and ferromagnetic cores are collectively known as Eddy current losses. Eddy current loss is essentially an I2R loss caused within a conductor caused resistance offered by the conductor to the flow of current. This loss is further increased by the temperature rise.

Hysteresis loss

When the alternating current changes its direction of flow, the direction of the magnetic field is also changed. This forces the molecules in the core to realign according to the direction. This movement cause friction and this loss is known as hysteresis loss.

Motor Sizing

Why is motor sizing important?

Motor sizing refers to the process of picking the correct motor for a given load.

It is important to size a motor correctlybecause:

- If a motor is too small for an application it may not have sufficient torque to start the load and run it up to the correct speed. Even if it does get the load up to speed the motor will overheat and burnout if it istoo smallfor the application.
- If a motor is too large for an application then money has been wasted in purchasing such a large motor. Also motors typically operate inefficiently when they are run well below rated load. So money is also wasted in running costs.

Choosing The Right Motor

For the load torque is constant then choosing amotor whose rated torque isslightly above the torque required by the load. The load torque should be between 75% - 100% of the rated motor toque with 95% being an ideal choice.

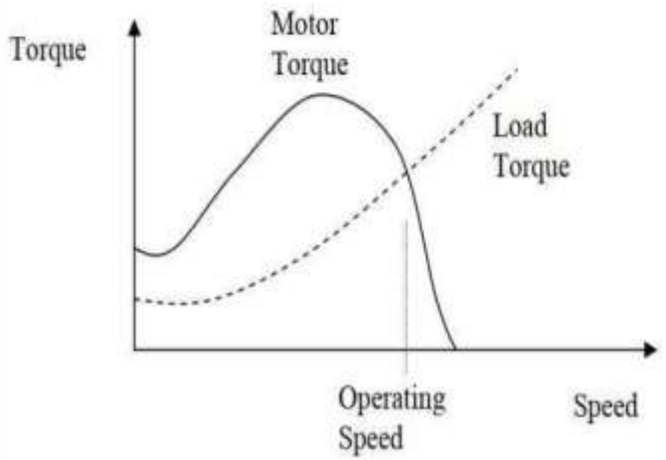
For constant speed applications where the motor will run continuously at rated speed you do not need to work out torque. Youcan simply look at load power and motor power and use the same 75%-100% rule. This is the case for standard induction motors without avariable speed drive.

Can The Motor Start The Load?

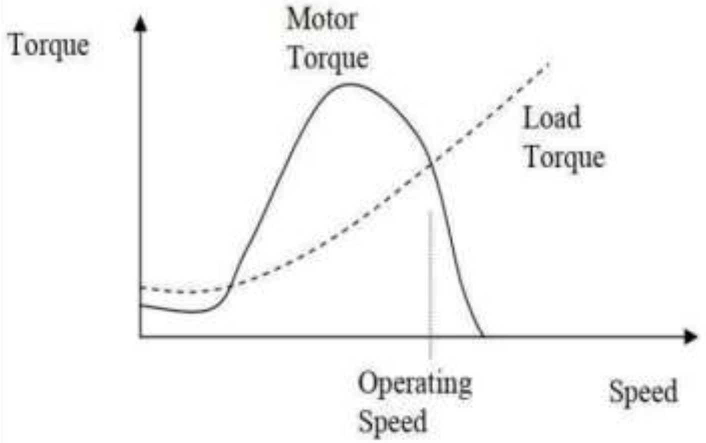
The torque produced by amotor varies with speed and the torque produced by a load also varies with speed.

- ➢ If the motor torque is greater than the load torque then the load will accelerate.
- ➢ If the load torque is greater than the motor torque then the load will decelerate.

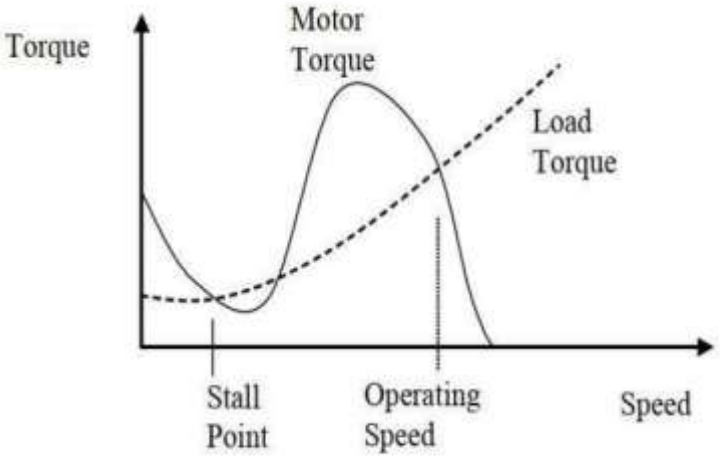
Accelerating Torque = T_{motor} -T_{load}



This motor will start the load and get it up to speed correctly.



This motor will never even start



This motor will start the load but it will get stuck at the stall point.

Motor Sizing for variable loads

Some loads do not present a constant torque even after they have got up to full speed.
This presents a variable power to the motor and complicates the sizing problem.

In general, we should ensure that -

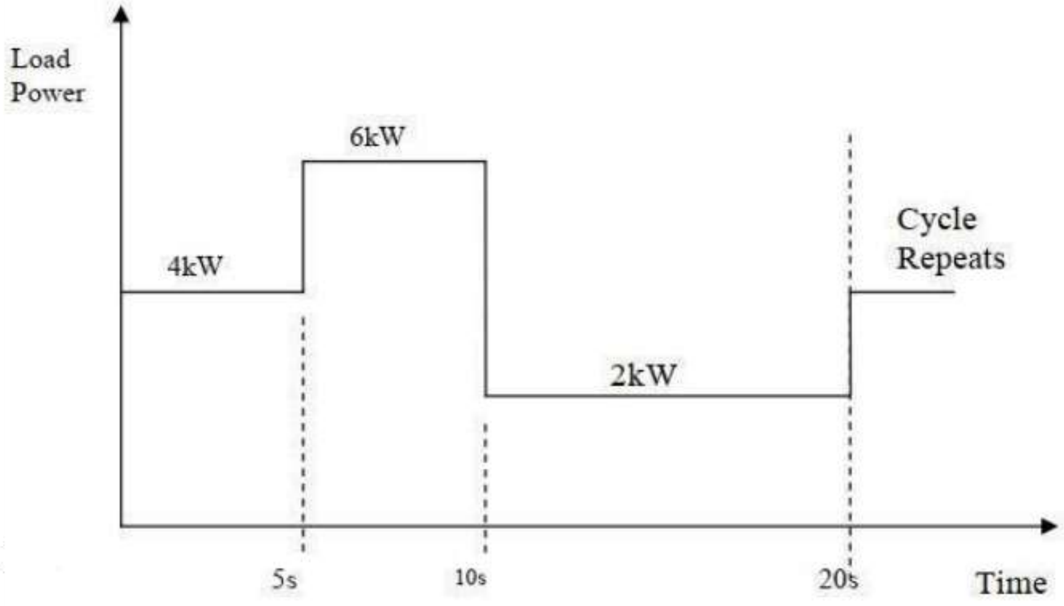
1. Peak load torque (or power) < Rated Motor torque (or power) x (1+Service factor/100%)

2. The root mean square load torque (power) requirement must be less than 100% of the rated motor torque (power) and ideally greater than 75% of rated motor torque (power).

3. Check that the Motor can start the load and get it up to speed.

Motor Sizing for variable power loads

Problem: The following load torque profile was measured for a constant speed load. Choose an appropriately sized motor to drive this load assuming a service factor of 20%. You do not need to consider starting performance.



i. Choose a suitable power rating of the motor/ Choose an appropriate size of the motor.
ii. Calculate the load power (rms).
iii. Is the calculated motor size acceptable?

i. Peak Load Power requirement = 6kW.
Peak load power must be less than
Rated Motor Power x ServiceFactor
=> 6kW < Pmotor x1.2 => Pmotor>6/1.2 =5kW

ii.

$$RMSLoadpower = \sqrt{\frac{P_1^2 t_1 + P_2^2 t_2 + P_3^2 t_3 + ... + P_n^2 t_n}{t_1 + t_2 + t_3 + ... + t_n}}$$

$$= \sqrt{\frac{4^2 \times 5 + 6^2 \times 5 + 2^2 \times 10}{5 + 5 + 10}} = 4.16kW$$

iii. If we choose Pmotor= 5kW then,
RMS load power = 4.16/5*100% = 83% of motor rating which is acceptable.
NB:If RMSload power turned out to be less than 75% of 5kW we sould still need to choose a 5kW motor. However if RMSload power was great than 100%of rated motor power we would need to choose a bigger motor.

Motor Duty Class

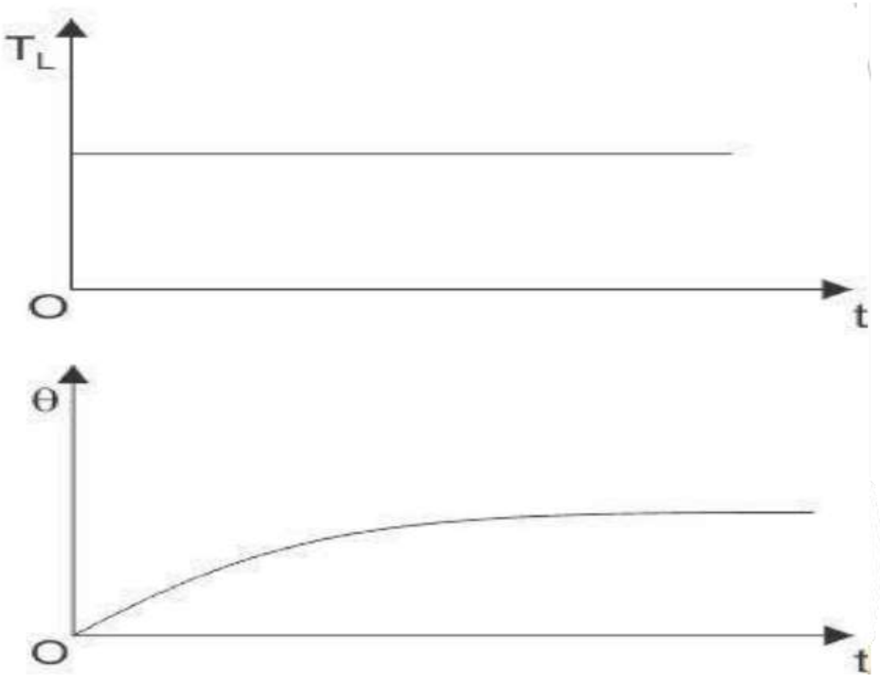
- The operating time for all motors are not the same. Some of the motors runs all the time, and some of the motor’s run time is shorter than the rest period. To control these motors electrical drives are employed in industries.
- The fraction of time in a cycle for which a motor stays ON is called the duty cycle. On the basis of this duty cycles there are 8 different motor duty classes:

1. Continuous duty
2. Short time duty
3. Intermittent periodic duty
4. Intermittent periodic duty with starting
5. Intermittent periodic duty with starting and braking
6. Continuous duty with intermittent periodic loading
7. Continuous duty with starting and braking
8. Continuous duty with periodic speed changes

Motor Duty Class

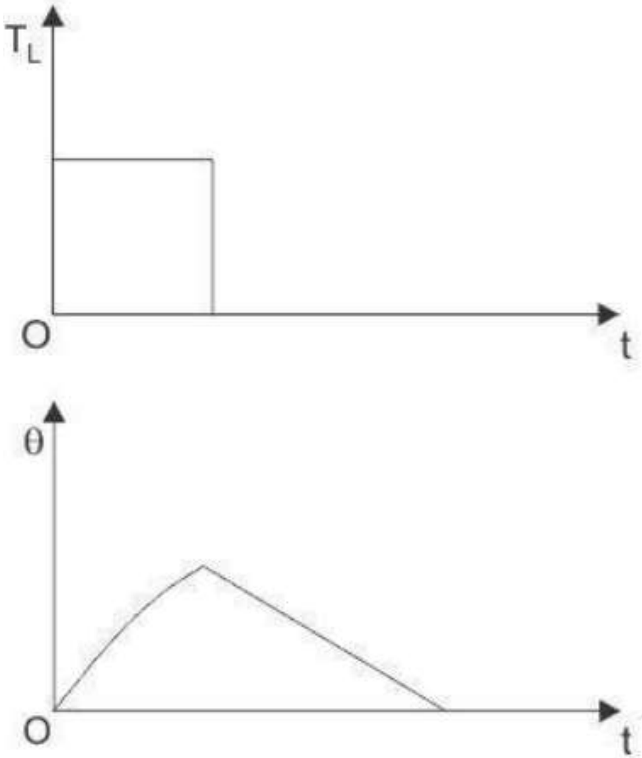
Continuous Duty Cycle

This duty denotes that, the motor is running long enough at constant torque and the electric motor temperature reaches the steady state value. Thesemotors are used in paper mill drives, compressors, conveyorsetc.



Short Time Duty Cycle

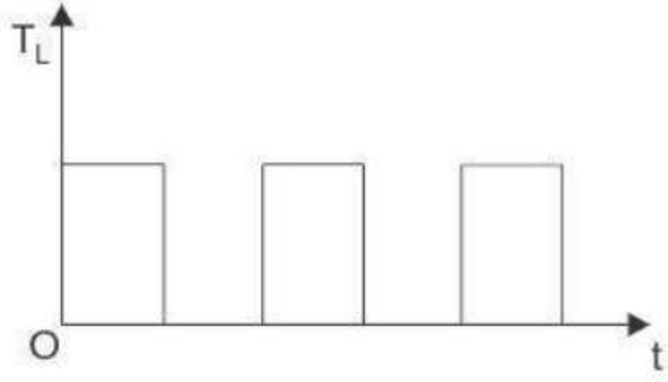
In these motors, the time of operation is very low and the heating time is much lower than the cooling time. So, the motor cools off to ambient temperature before operating again. These motors are used in crane drives, drives for house hold appliances, valve drives etc.

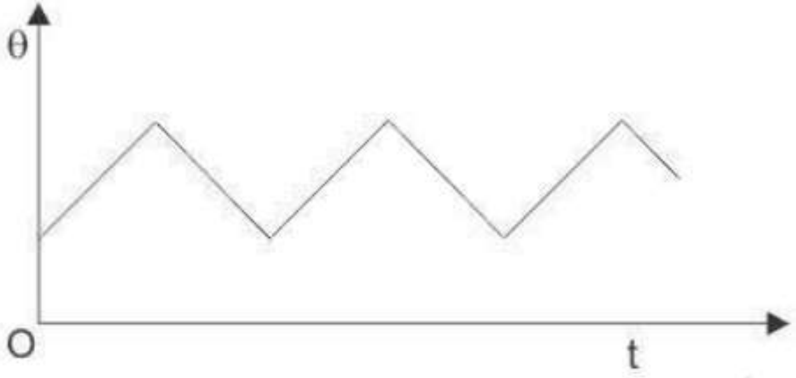


Motor Duty Class

Intermittent Periodic Duty

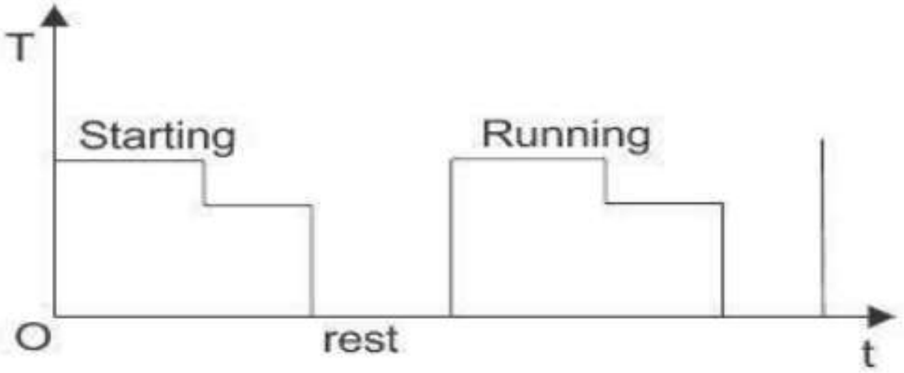
Here the motor operates for some time and then there is rest period. In both cases, the time is insufficient to raise the temperature to steady state value or cool it off to ambient temperature. This is seen at press and drilling machine drive

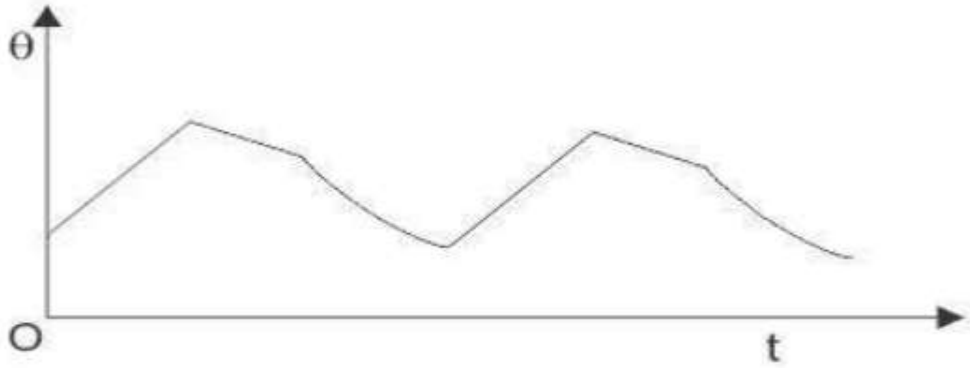




Intermittent Periodic Duty With Starting

In this type of duty, there is a period of starting, which can not be ignored and there is a heat loss at that time. After that there is running period and rest period which are not adequate to attain the steady state temperatures. This motor duty class is widely used in metal cutting and drilling tool drives, mine hoist etc.

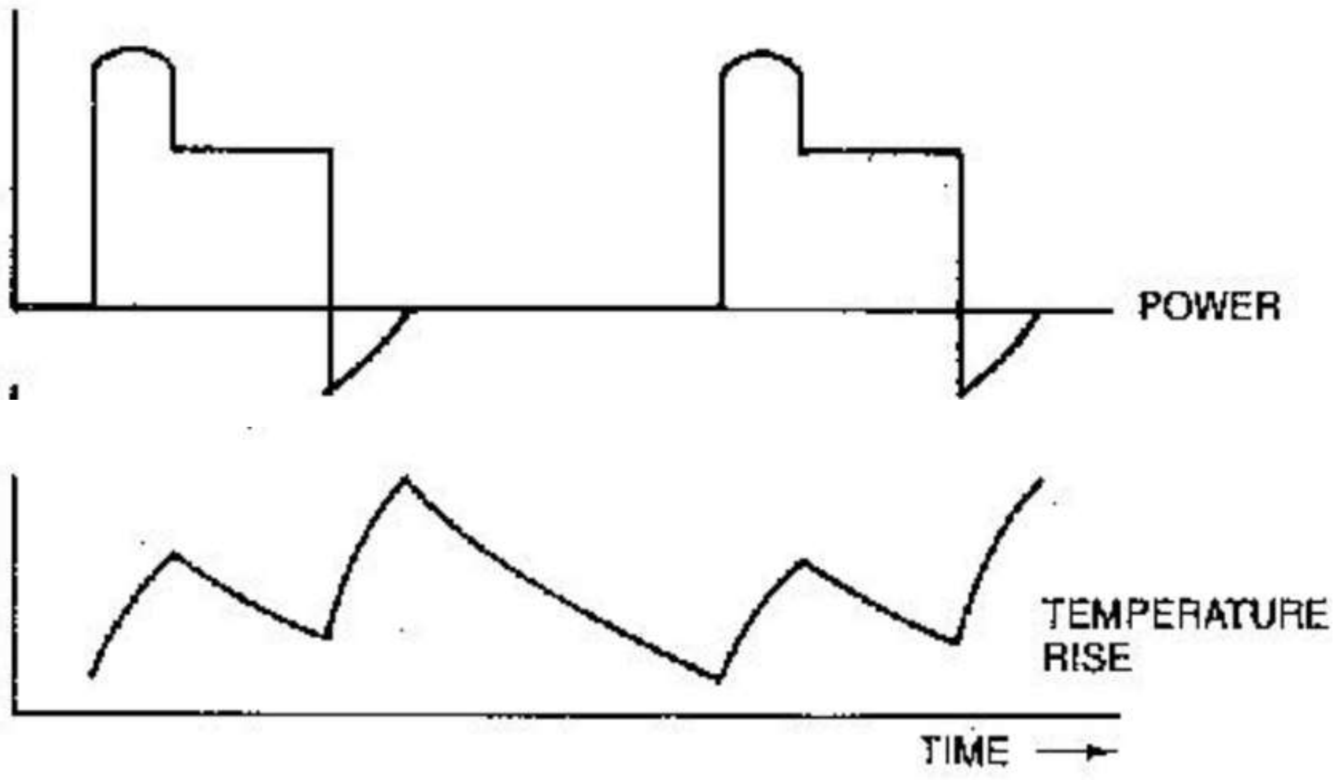




Motor Duty Class

Intermittent Periodic Duty Starting & Breaking

In this type of drives, heat loss during starting and braking cannot be ignored. So, the corresponding periods are starting period, operating period, braking period and resting period, but all the periods are too short to attain the respective steady state temperatures, these techniques are used in billet mill drive, manipulator drive, mine hoist etc



Motor Duty Class

Continuous Duty with Intermittent Periodic Loading:
In this type of motor duty, everything is same as the periodic duty but here a no load running period occurs instead of the rest period. Pressing, cutting are the examplesof this system.
Continuous Duty with Starting and Braking:
It is also a period of starting, running and braking and there is no resting period. Themain drive of ablooming mill is an example.
Continuous Duty with Periodic Speed Changes:
In this type of motor duty, there are different running periods at different loads and speeds. But there is no rest period and all the periods are too short to attain the steady-state temperatures.

Motor De-rating

Motor De –rating means intentionally reducing a motor's maximum power output, torque or speed to run safely. Any adverse operating conditions require that the motor performance be de-rated.

These conditions include -

- ambient temperature above 40°C,
- motor mounting position: Mounting Arrangement: The motor torque must be de-rated if the motor mounting surface is heated from an external source, such as a gearbox.
- drive switching frequency
- the drive being oversized for the motor

Energy Efficient Motor

- Energy efficient electric motors utilize improved motor design.
- It uses high quality materials such as thin Si lamination to reduce motor losses.
- Energy efficient motors are designed to reduce power loss and improve efficiency by 4 to 6%.
- It uses less electricity, run cooler and often last longer than **NEMA B** motors of the samesize.
- The improved design results in less heat dissipation and reduced noise output.